

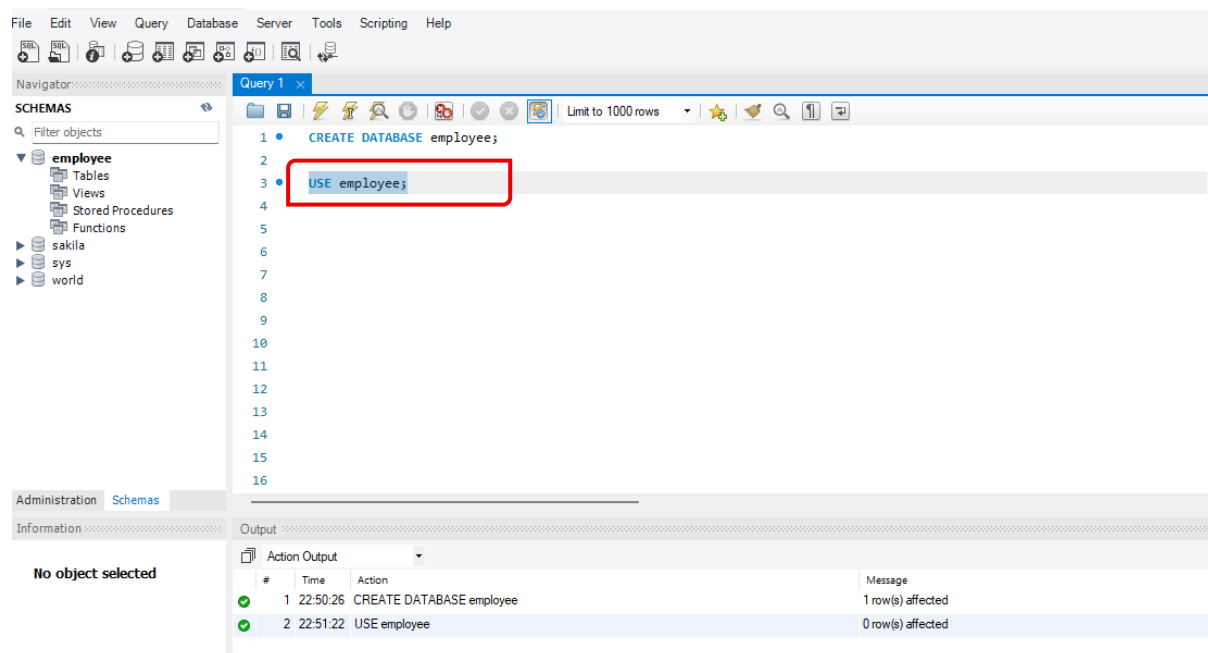
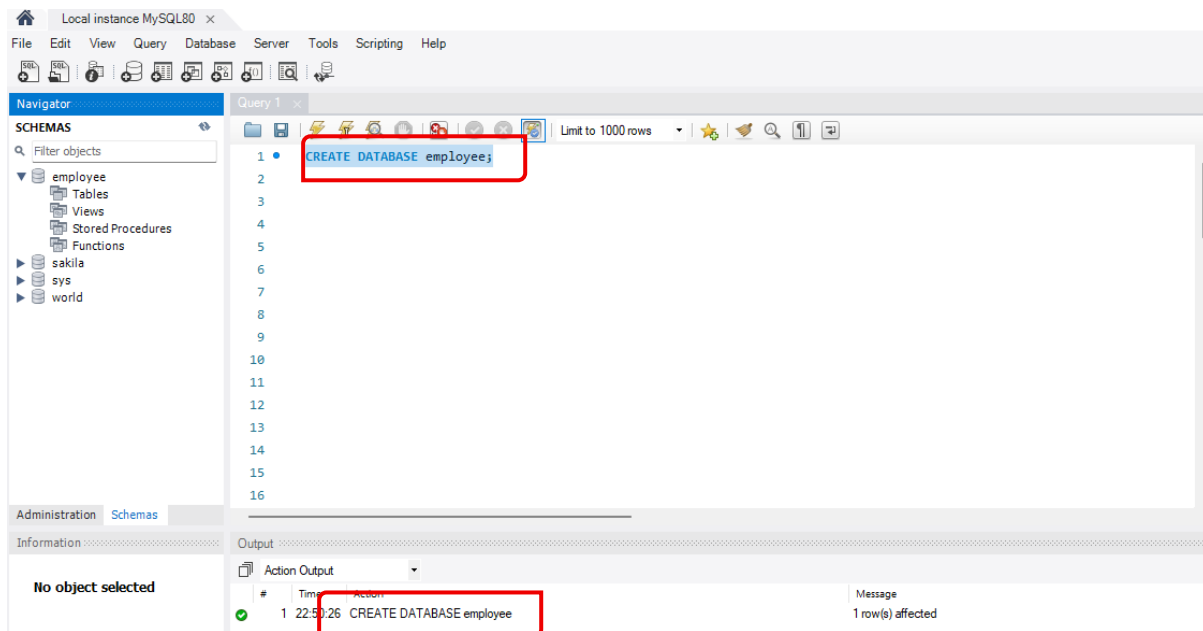
DDL Commands, Constraints

OBJECTIVE:

The objective of this assignment is to create and modify an Employee Database using MySQL DDL commands and implement appropriate constraints to maintain data integrity.

1.Database Creation:

The employee database was created using the CREATE DATABASE command. The database was then selected using the USE command.

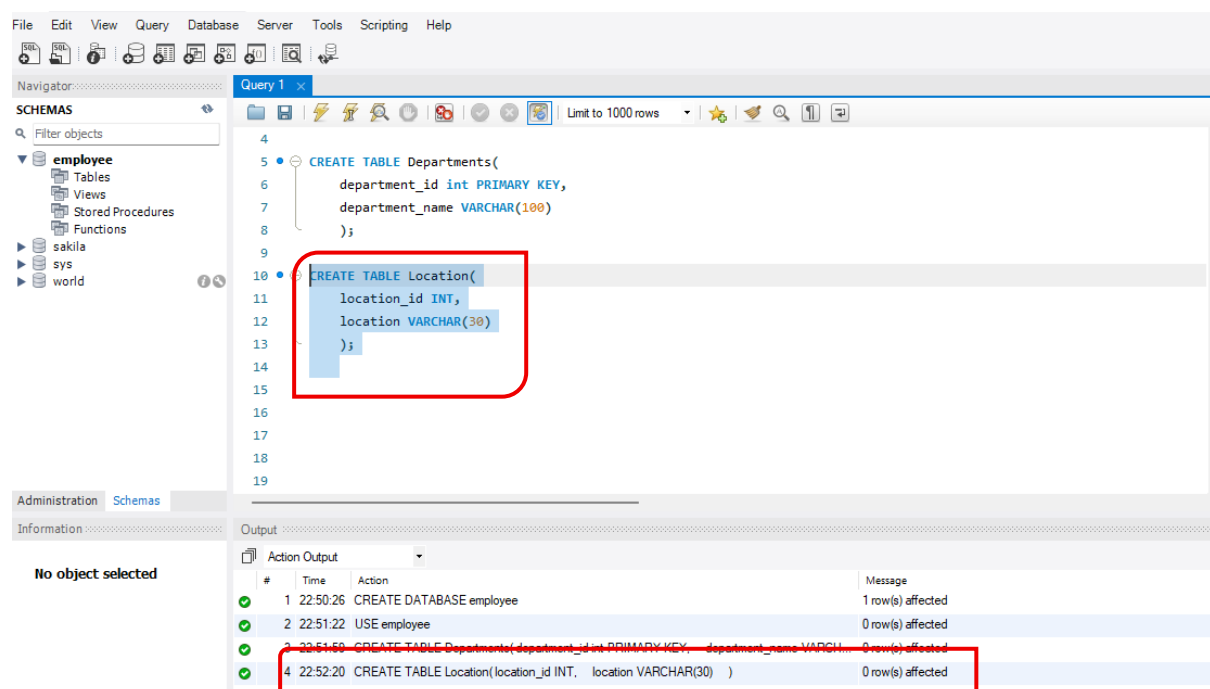
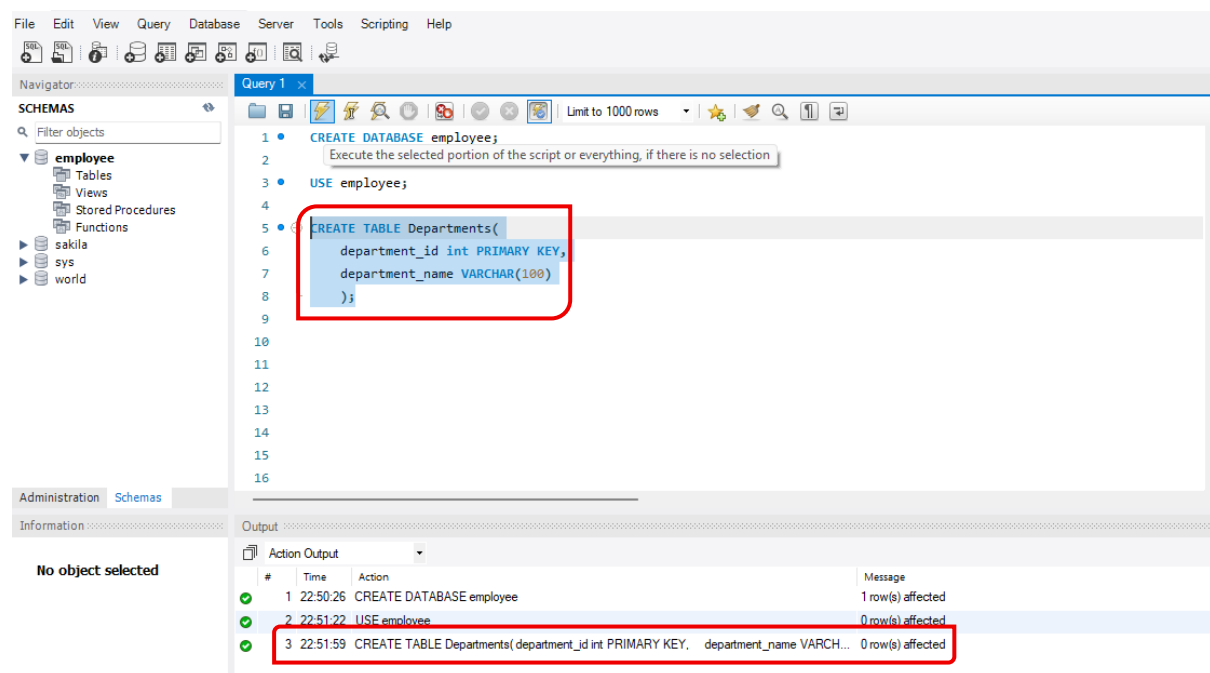


2. Table Creation

Three tables were created based on the given schema:

- Departments
- Location
- Employees

The tables were created using the CREATE TABLE command.



File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- employee
 - Tables
 - Views
 - Stored Procedures
 - Functions
- sakila
- sys
- world

Query 1

```

10 CREATE TABLE Location(
11     location_id INT,
12     location VARCHAR(30)
13 );
14
15 CREATE TABLE employees(
16     employee_id INT PRIMARY KEY,
17     employee_name VARCHAR(50),
18     gender enum('Female','Male'),
19     age INT,
20     hire_date DATE,
21     designation VARCHAR(100),
22     department_id INT,
23     location_id INT,
24     salary DECIMAL(10,2)
25 );

```

Administration Schemas

Information

No object selected

Output

Action Output

#	Time	Action	Message
1	22:50:26	CREATE DATABASE employee	1 row(s) affected
2	22:51:22	USE employee	0 row(s) affected
3	22:51:59	CREATE TABLE Departments(department_id int PRIMARY KEY, department_name VARCHAR...	0 row(s) affected
4	22:52:20	CREATE TABLE Location(location_id INT, location VARCHAR(30))	0 row(s) affected
5	22:52:50	CREATE TABLE employees(employee_id INT PRIMARY KEY, employee_name VARCHAR(...	0 row(s) affected

Inserted some values to check the integrity .

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- employee
 - Tables
 - Views
 - Stored Procedures
 - Functions
- sakila
- sys
- world

Query 1

```

19 age INT,
20 hire_date DATE,
21 designation VARCHAR(100),
22 department_id INT,
23 location_id INT,
24 salary DECIMAL(10,2)
25 );
26
27 INSERT INTO employees
28 VALUES (100, 'SHA','Female',14,'2008-07-21','IT',1001,400,12000.57);
29
30
31
32
33
34

```

Administration Schemas

Information

No object selected

Output

Action Output

#	Time	Action	Message
1	22:50:26	CREATE DATABASE employee	1 row(s) affected
2	22:51:22	USE employee	0 row(s) affected
3	22:51:59	CREATE TABLE Departments(department_id int PRIMARY KEY, department_name VARCHAR...	0 row(s) affected
4	22:52:20	CREATE TABLE Location(location_id INT, location VARCHAR(30))	0 row(s) affected
5	22:52:50	CREATE TABLE employees(employee_id INT PRIMARY KEY, employee_name VARCHAR(...	0 row(s) affected
6	22:54:39	INSERT INTO employees VALUES (100, 'SHA','Female',14,'2008-07-21','IT',1001,400,12000.57);	1 row(s) affected

Object Info Session

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Query 1 x

Limit to 1000 rows

```

23     location_id INT,
24     salary DECIMAL(10,2)
25 );
26
27 • INSERT INTO employees
28   VALUES (100, 'SHA', 'Female', 14, '2008-07-21', 'IT', 1001, 400, 12000.57);
29
30 • SELECT * FROM employees;
31

```

Result Grid

employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
100	SHA	Female	14	2008-07-21	IT	1001	400	12000.57

Administration Schemas

Information: No object selected

Object Info Session

employees 6 x

Output

Action Output

#	Time	Action	Message
2	22:51:22	USE employee	0 row(s) affected
3	22:51:59	CREATE TABLE Departments(department_id int PRIMARY KEY, department_name VAR...	0 row(s) affected
4	22:52:20	CREATE TABLE Location(location_id INT, location VARCHAR(30))	0 row(s) affected
5	22:52:50	CREATE TABLE employees(employee_id INT PRIMARY KEY, employee_name VARCHA...	0 row(s) affected
6	22:54:39	INSERT INTO employees VALUES (100, 'SHA', 'Female', 14, '2008-07-21', 'IT', 1001, 400, 120...	1 row(s) affected
7	22:55:14	SELECT * FROM employees LIMIT 0, 1000	1 row(s) returned

3. Table Alteration (ALTER):

➊ Added a new column named "email" to the Employees table to store employee email addresses.

File Edit View Query Database Server Tools Scripting Help

Navigator: Query 1 x

Limit to 1000 rows

```

26
27 • INSERT INTO employees
28   VALUES (100, 'SHA', 'Female', 14, '2008-07-21', 'IT', 1001, 400, 12000.57);
29
30 • SELECT * FROM employees;
31
32 • ALTER TABLE employees ADD email VARCHAR(100);
33 • SELECT * FROM employees;
34

```

Result Grid

employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary	email
100	SHA	Female	14	2008-07-21	IT	1001	400	12000.57	

Administration Schemas

Information: No object selected

Object Info Session

employees 7 x

Output

Action Output

#	Time	Action	Message
5	22:52:50	CREATE TABLE employees(employee_id INT PRIMARY KEY, employee_name VARCHA...	0 row(s) affected
6	22:54:39	INSERT INTO employees VALUES (100, 'SHA', 'Female', 14, '2008-07-21', 'IT', 1001, 400, 120...	1 row(s) affected
7	22:55:14	SELECT * FROM employees LIMIT 0, 1000	1 row(s) returned
8	22:56:53	ALTER TABLE employees ADD email VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
9	22:57:21	ALTER TABLE employees ADD email VARCHAR(100)	Error Code: 1060. Duplicate column name 'email'
10	22:57:31	SELECT * FROM employees LIMIT 0, 1000	1 row(s) returned

➋ Modified the data type of the "designation" column in the Employees table to support a wider range of values.

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- employee
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 - Views
 - Stored Procedures
 - Functions
- sakila
- sys
- world

Query 1

```

26
27 • INSERT INTO employees
28   VALUES (100, 'SHA', 'Female', 14, '2008-07-21', 'IT', 1001, 400, 12000.57);
29
30 • SELECT * FROM employees;
31
32 • ALTER TABLE employees ADD email VARCHAR(100);
33 • SELECT * FROM employees;
34
35 • ALTER TABLE employees MODIFY COLUMN designation LONGTEXT;
36
37
38
39
40
41

```

Administration Schemas

Information

No object selected

Object Info Session

Output

Action Output

#	Time	Action	Message
6	22:54:39	INSERT INTO employees VALUES (100, 'SHA', 'Female', 14, '2008-07-21', 'IT', 1001, 400, 12000.57);	1 row(s) affected
7	22:55:14	SELECT * FROM employees LIMIT 0, 1000	1 row(s) returned
8	22:56:53	ALTER TABLE employees ADD email VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
9	22:57:21	ALTER TABLE employees ADD email VARCHAR(100)	Error Code: 1060. Duplicate column name 'email'
10	22:57:31	SELECT * FROM employees LIMIT 0, 1000	1 row(s) returned
11	22:58:19	ALTER TABLE employees MODIFY COLUMN designation LONGTEXT	1 row(s) affected Records: 1 Duplicates: 0 Warnings: 0

● Dropped the “age” column from the Employees table.

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- employee
 - Tables
 - Views
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 - Functions
- sakila
- sys
- world

Query 1

```

26
27 • INSERT INTO employees
28   VALUES (100, 'SHA', 'Female', 14, '2008-07-21', 'IT', 1001, 400, 12000.57);
29
30 • SELECT * FROM employees;
31
32 • ALTER TABLE employees ADD email VARCHAR(100);
33 • SELECT * FROM employees;
34 • ALTER TABLE employees MODIFY COLUMN designation LONGTEXT;
35 • ALTER TABLE employees DROP COLUMN age;
36
37
38
39
40
41

```

Administration Schemas

Information

No object selected

Object Info Session

Output

Action Output

#	Time	Action	Message
7	22:55:14	SELECT * FROM employees LIMIT 0, 1000	1 row(s) returned
8	22:56:53	ALTER TABLE employees ADD email VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
9	22:57:21	ALTER TABLE employees ADD email VARCHAR(100)	Error Code: 1060. Duplicate column name 'email'
10	22:57:31	SELECT * FROM employees LIMIT 0, 1000	1 row(s) returned
11	22:58:19	ALTER TABLE employees MODIFY COLUMN designation LONGTEXT	1 row(s) affected Records: 1 Duplicates: 0 Warnings: 0
12	22:58:57	ALTER TABLE employees DROP COLUMN age	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

② Renamed the "hire_date" column to "date_of_joining".

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with the 'employee' schema selected. The main editor shows a SQL script with the following queries:

```
26
27 • IN Execute the selected portion of the script or everything, if there is no selection
28 VALUES (100, 'SHA', 'Female', 14, '2008-07-21', 'IT', 1001, 400, 12000.57);
29
30 • SELECT * FROM employees;
31
32 • ALTER TABLE employees ADD email VARCHAR(100);
33 • SELECT * FROM employees;
34 • ALTER TABLE employees MODIFY COLUMN designation LONGTEXT;
35 • ALTER TABLE employees DROP COLUMN age;
36 • ALTER TABLE employees RENAME COLUMN hire_date TO date_of_joining;
37
38 • ALTER TABLE Location
39 RENAME TO Locations;
40
41
```

The query on line 36 is highlighted with a red box. The bottom 'Output' pane shows the execution results:

#	Time	Action	Message
9	22:57:21	ALTER TABLE employees ADD email VARCHAR(100)	Error Code: 1060. Duplicate column name 'email'
10	22:57:31	SELECT * FROM employees LIMIT 0, 1000	1 row(s) returned
11	22:58:19	ALTER TABLE employees MODIFY COLUMN designation LONGTEXT	1 row(s) affected Records: 1 Duplicates: 0 Warnings: 0
12	22:58:57	ALTER TABLE employees DROP COLUMN age	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
13	23:01:45	ALTER TABLE Location RENAME TO Locations	0 row(s) affected
14	23:02:33	ALTER TABLE employees RENAME COLUMN hire_date TO date_of_joining	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

The output for line 14 is also highlighted with a red box.

3. Table Renaming (RENAME):

② Renamed the "Departments" table to "Departments_Info".

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with the 'employee' schema selected. The main editor shows a SQL script with the following queries:

```
29
30 • SELECT * FROM employees;
31
32 • ALTER TABLE employees ADD email VARCHAR(100);
33 • SELECT * FROM employees;
34 • ALTER TABLE employees MODIFY COLUMN designation LONGTEXT;
35 • ALTER TABLE employees DROP COLUMN age;
36 • ALTER TABLE employees RENAME COLUMN hire_date TO date_of_joining;
37
38 • ALTER TABLE Location
39 RENAME TO Locations;
40
41 • ALTER TABLE Departments
42 RENAME TO Departments_info;
43
44
```

The queries on lines 41 and 42 are highlighted with a red box. The bottom 'Output' pane shows the execution results:

#	Time	Action	Message
10	22:57:31	SELECT * FROM employees LIMIT 0, 1000	1 row(s) returned
11	22:58:19	ALTER TABLE employees MODIFY COLUMN designation LONGTEXT	1 row(s) affected Records: 1 Duplicates: 0 Warnings: 0
12	22:58:57	ALTER TABLE employees DROP COLUMN age	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
13	23:01:45	ALTER TABLE Location RENAME TO Locations	0 row(s) affected
14	23:02:33	ALTER TABLE employees RENAME COLUMN hire_date TO date_of_joining	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
15	23:03:39	ALTER TABLE Departments RENAME TO Departments_info	0 row(s) affected

The output for line 15 is also highlighted with a red box.

● Renamed the "Location" table to "Locations".

The screenshot shows the MySQL Workbench interface. The 'SCHEMAS' pane on the left shows the 'employee' schema with tables 'departments', 'employees', and 'location'. The 'Query 1' editor contains the following SQL script:

```
26
27 • INSERT INTO employees
28   VALUES (100, 'SHA', 'Female', 14, '2008-07-21', 'IT', 1001, 400, 12000.57);
29
30 • SELECT * FROM employees;
31
32 • ALTER TABLE employees ADD email VARCHAR(100);
33 • SELECT * FROM employees;
34 • ALTER TABLE employees MODIFY COLUMN designation LONGTEXT;
35 • ALTER TABLE employees DROP COLUMN age;
36 • ALTER TABLE employees RENAME COLUMN hire_date TO date_of_joining;
37
38 • ALTER TABLE Location
39   RENAME TO Locations;
40
41
```

The 'Output' pane at the bottom shows the execution results:

#	Time	Action	Message
8	22:56:53	ALTER TABLE employees ADD email VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
9	22:57:21	ALTER TABLE employees ADD email VARCHAR(100)	Error Code: 1060. Duplicate column name 'email'
10	22:57:31	SELECT * FROM employees LIMIT 0, 1000	1 row(s) returned
11	22:58:19	ALTER TABLE employees MODIFY COLUMN designation LONGTEXT	1 row(s) affected Records: 1 Duplicates: 0 Warnings: 0
12	22:58:57	ALTER TABLE employees DROP COLUMN age	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
13	23:01:45	ALTER TABLE Location RENAME TO Locations	0 row(s) affected

4. Table Truncation (TRUNCATE): Truncated the Employees table.

The screenshot shows the MySQL Workbench interface. The 'SCHEMAS' pane on the left shows the 'employee' schema with tables 'departments', 'employees', and 'location'. The 'Query 1' editor contains the following SQL script:

```
34 • ALTER TABLE employees MODIFY COLUMN designation LONGTEXT;
35 • ALTER TABLE employees DROP COLUMN age;
36 • ALTER TABLE employees RENAME COLUMN hire_date TO date_of_joining;
37
38 • ALTER TABLE Location
39   RENAME TO Locations;
40
41 • ALTER TABLE Departments
42   RENAME TO Departments_info;
43
44 • TRUNCATE TABLE employees;
45
46
47
48
49
```

The 'Output' pane at the bottom shows the execution results:

#	Time	Action	Message
11	22:58:19	ALTER TABLE employees MODIFY COLUMN designation LONGTEXT	1 row(s) affected Records: 1 Duplicates: 0 Warnings: 0
12	22:58:57	ALTER TABLE employees DROP COLUMN age	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
13	23:01:45	ALTER TABLE Location RENAME TO Locations	0 row(s) affected
14	23:02:33	ALTER TABLE employees RENAME COLUMN hire_date TO date_of_joining	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
15	23:02:39	ALTER TABLE Departments RENAME TO Departments_info	0 row(s) affected
16	23:04:14	TRUNCATE TABLE employees	0 row(s) affected

5. Database & Table Dropping (DROP): Dropped the Employees table and then the “employee” database.

The screenshot shows the SQL Enterprise Manager interface. The 'Query 1' script contains the following SQL statements:

```
34 • ALTER TABLE employees MODIFY COLUMN designation LONGTEXT;
35 • ALTER TABLE employees DROP COLUMN age;
36 • ALTER TABLE employees RENAME COLUMN hire_date TO date_of_joining;
37
38 • ALTER TABLE Location
39   RENAME TO Locations;
40
41 • ALTER TABLE Departments
42   RENAME TO Departments_info;
43
44 • TRUNCATE TABLE employees;
45
46 • DROP TABLE employees;
47 • DROP DATABASE employee;
48
49
```

The 'Output' pane shows the execution results:

#	Time	Action	Message
13	23:01:45	ALTER TABLE Location RENAME TO Locations	0 row(s) affected
14	23:02:33	ALTER TABLE employees RENAME COLUMN hire_date TO date_of_joining	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
15	23:03:39	ALTER TABLE Departments RENAME TO Departments_info	0 row(s) affected
16	23:04:14	TRUNCATE TABLE employees	0 row(s) affected
17	23:05:28	DROP TABLE employees	0 row(s) affected
18	23:05:28	DROP DATABASE employee	2 row(s) affected

Constraints :

1. Database Recreation:

● Dropped the 'employee' database if it exists

The screenshot shows the SQL Enterprise Manager interface. The 'Query 1' script contains the following SQL statements:

```
41 • ALTER TABLE Departments
42   RENAME TO Departments_info;
43
44 • TRUNCATE TABLE employees;
45
46 • DROP TABLE employees;
47 • DROP DATABASE employee;
48
49 -- Drop the 'employee' database if it exists
50
51 • DROP DATABASE IF EXISTS employee;
52
53
54
55
56
```

The 'Output' pane shows the execution results:

#	Time	Action	Message
14	23:02:33	ALTER TABLE employees RENAME COLUMN hire_date TO date_of_joining	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
15	23:03:39	ALTER TABLE Departments RENAME TO Departments_info	0 row(s) affected
16	23:04:14	TRUNCATE TABLE employees	0 row(s) affected
17	23:05:28	DROP TABLE employees	0 row(s) affected
18	23:05:28	DROP DATABASE employee	2 row(s) affected
19	23:13:16	DROP DATABASE IF EXISTS employee	0 row(s) affected, 1 warning(s): 1008 Can't drop database 'employee'; database doesn't exist

recreated it using the provided schema, ensuring that all tables are created with the appropriate constraints as instructed.

Query 1

```
46 DROP TABLE employees;
47 DROP DATABASE employee;
48
49 -- Drop the 'employee' database if it exists
50
51 DROP DATABASE IF EXISTS employee;
52
53 CREATE DATABASE employee;
54
55
56
57
58
59
60
61
```

Output

#	Time	Action	Message
15	23:03:39	ALTER TABLE Departments RENAME TO Departments_info	0 row(s) affected
16	23:04:14	TRUNCATE TABLE employees	0 row(s) affected
17	23:05:28	DROP TABLE employees	0 row(s) affected
18	23:05:28	DROP DATABASE employee	2 row(s) affected
19	23:13:16	DROP DATABASE IF EXISTS employee	0 row(s) affected, 1 warning(s): 1008 Can't drop database 'employee'; database exists
20	23:14:30	CREATE DATABASE employee	1 row(s) affected

2. Departments Table:

- Ensured that the "department_id" uniquely identifies each department.
- Set up constraints on the "department_name" to avoid duplicate and null entries.

Query 1

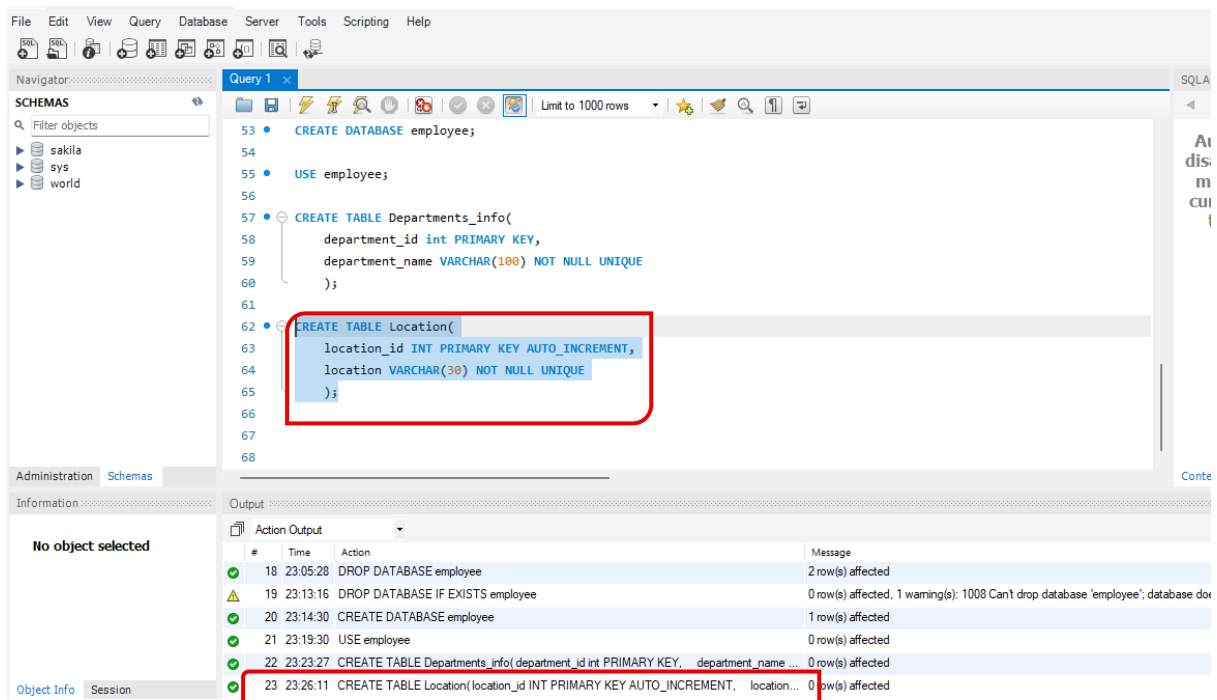
```
50
51 DROP DATABASE IF EXISTS employee;
52
53 CREATE DATABASE employee;
54
55 USE employee;
56
57 CREATE TABLE Departments_info(
58     department_id int PRIMARY KEY,
59     department_name VARCHAR(100) NOT NULL UNIQUE
60 );
61
62
63
64
65
```

Output

#	Time	Action	Message
17	23:05:28	DROP TABLE employees	0 row(s) affected
18	23:05:28	DROP DATABASE employee	2 row(s) affected
19	23:13:16	DROP DATABASE IF EXISTS employee	0 row(s) affected, 1 warning(s): 1008 Can't drop database 'employee'; database exists
20	23:14:30	CREATE DATABASE employee	1 row(s) affected
21	23:19:30	USE employee	0 row(s) affected
22	23:23:27	CREATE TABLE Departments_info(department_id int PRIMARY KEY, department_name ...	0 row(s) affected

3. Location Table:

- Established a mechanism to automatically generate unique identifiers for each location, ensuring that they are incremented sequentially.
- Implemented constraints to prevent the insertion of null and duplicate locations.



```
53 • CREATE DATABASE employee;
54
55 • USE employee;
56
57 • CREATE TABLE Departments_info(
58     department_id int PRIMARY KEY,
59     department_name VARCHAR(100) NOT NULL UNIQUE
60 );
61
62 • CREATE TABLE Location(
63     location_id INT PRIMARY KEY AUTO_INCREMENT,
64     location VARCHAR(30) NOT NULL UNIQUE
65 );
66
67
68
```

#	Time	Action	Message
18	23:05:28	DROP DATABASE employee	2 row(s) affected
19	23:13:16	DROP DATABASE IF EXISTS employee	0 row(s) affected, 1 warning(s): 1008 Can't drop database 'employee'; database does not exist
20	23:14:30	CREATE DATABASE employee	1 row(s) affected
21	23:19:30	USE employee	0 row(s) affected
22	23:23:27	CREATE TABLE Departments_info(department_id int PRIMARY KEY, department_name ...	0 row(s) affected
23	23:26:11	CREATE TABLE Location(location_id INT PRIMARY KEY AUTO_INCREMENT, location...	0 row(s) affected

4. Employees Table:

- Guaranteed that each employee has a distinct identifier.
- Created a restriction to ensure that the employee's name is always provided.
- Limited the acceptable values for the "gender" field to only 'M' or 'F'.
- Enforced a condition to ensure that the employee's age is 18 or above.
- Automatically assigned the current date to the "hire_date" field if not specified.
- Established links between the "department_id" and "location_id" fields in the "employees" table and their respective tables.

File Edit View Query Database Server Tools Scripting Help

Navigator: Query 1 x

SCHMAS

Filter objects

sakila
 sys
 world

Execute the selected portion of the script or everything, if there is no selection

```
64  
65  
66  
67 • CREATE TABLE employees(  
68     employee_id INT PRIMARY KEY,  
69     employee_name VARCHAR(50) NOT NULL,  
70     gender enum('M','F'),  
71     age INT CHECK(age>=18),  
72     hire_date DATE DEFAULT(current_date),  
73     designation VARCHAR(100),  
74     department_id INT,  
75     location_id INT,  
76     salary DECIMAL(10,2),  
77     FOREIGN KEY (department_id) REFERENCES Departments_info (department_id),  
78     FOREIGN KEY (location_id) REFERENCES Location (location_id)  
79 );
```

Administration Schemas

Information

No object selected

Object Info Session

Output

Action Output

#	Time	Action	Message
19	23:13:16	DROP DATABASE IF EXISTS employee	0 row(s) affected, 1 warning(s): 1008 Can't drop database 'employee'; dat
20	23:14:30	CREATE DATABASE employee	1 row(s) affected
21	23:19:30	USE employee	0 row(s) affected
22	23:23:27	CREATE TABLE Departments_info(department_id int PRIMARY KEY, department_name ...	0 row(s) affected
23	23:26:11	CREATE TABLE Location(location_id INT PRIMARY KEY AUTO_INCREMENT, location...	0 row(s) affected
24	23:38:11	CREATE TABLE employees(employee_id INT PRIMARY KEY, employee_name VARCHA...	0 row(s) affected